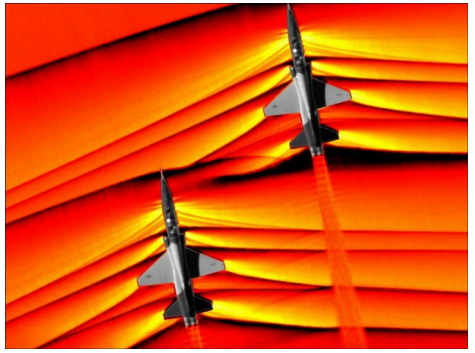
Machine Learning for Shock-Capturing Numerical Schemes

WENO-NN, NN-TENO5 and Differentiable CFD for 1D Hyperbolic Conservation Laws

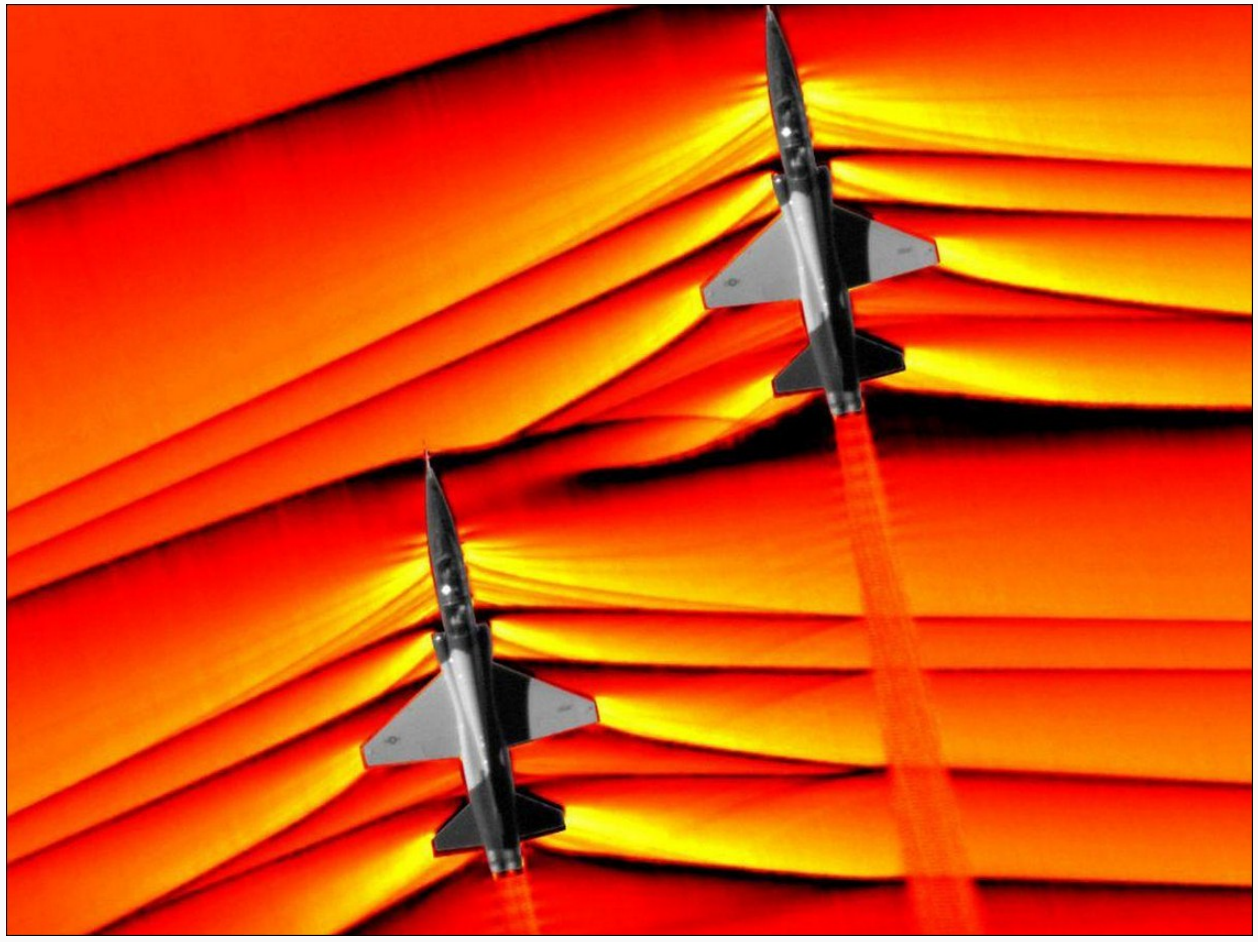
Yasser Baouzil

Supervisor: Radouan Boukharfane

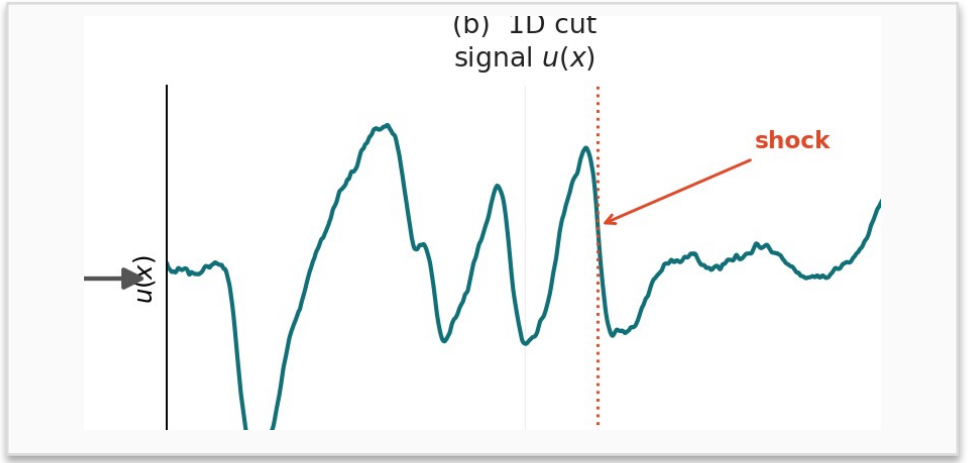
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Physical motivation: conservation laws describe flows with shocks



Real shock-wave photograph: visible compressible-flow structures and abrupt changes.



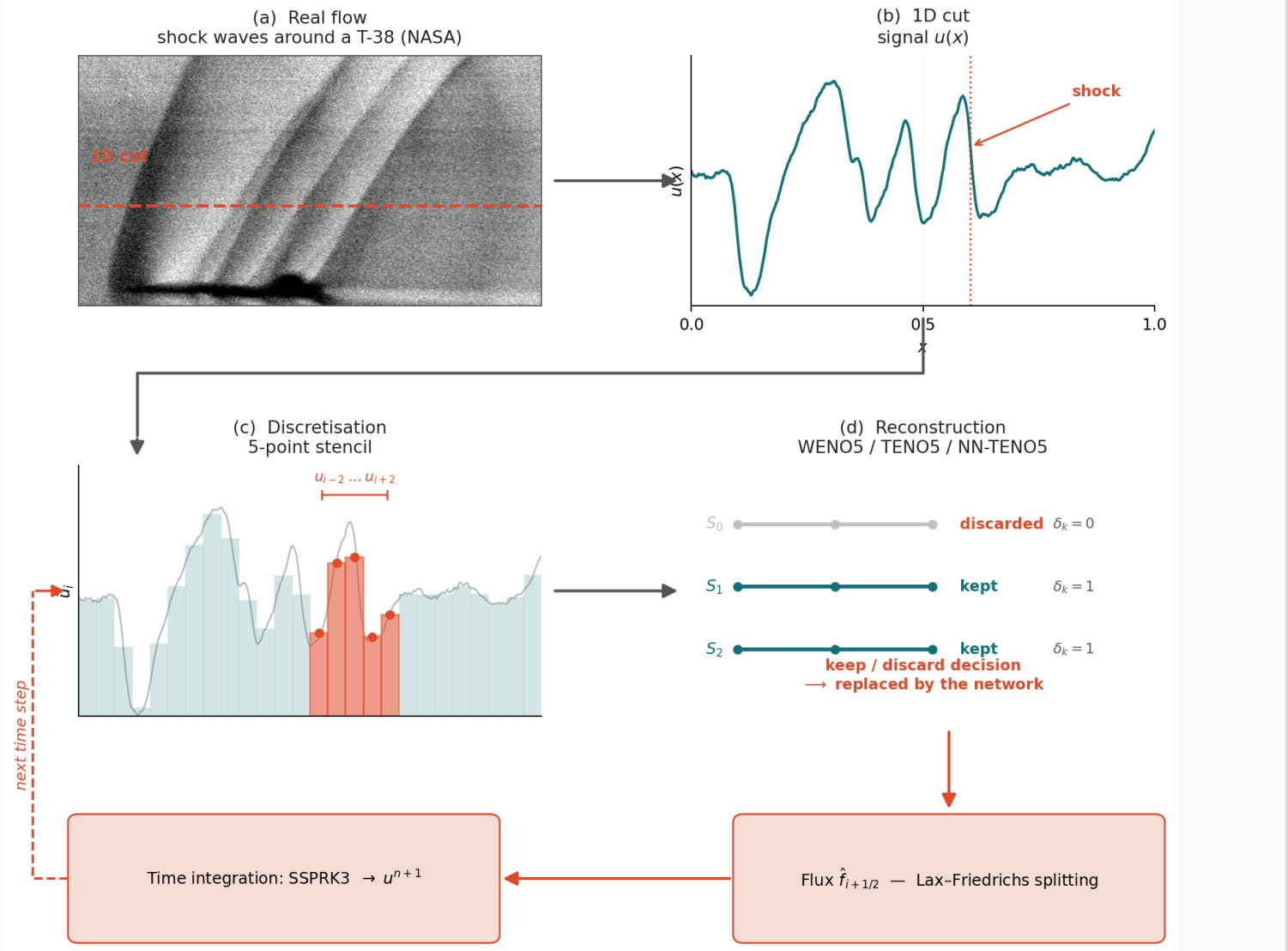
$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0$$

Shocks are discontinuities in the solution.

Conservation laws are the starting point.

The numerical method must avoid spurious oscillations.

From conservation laws to a numerical solver



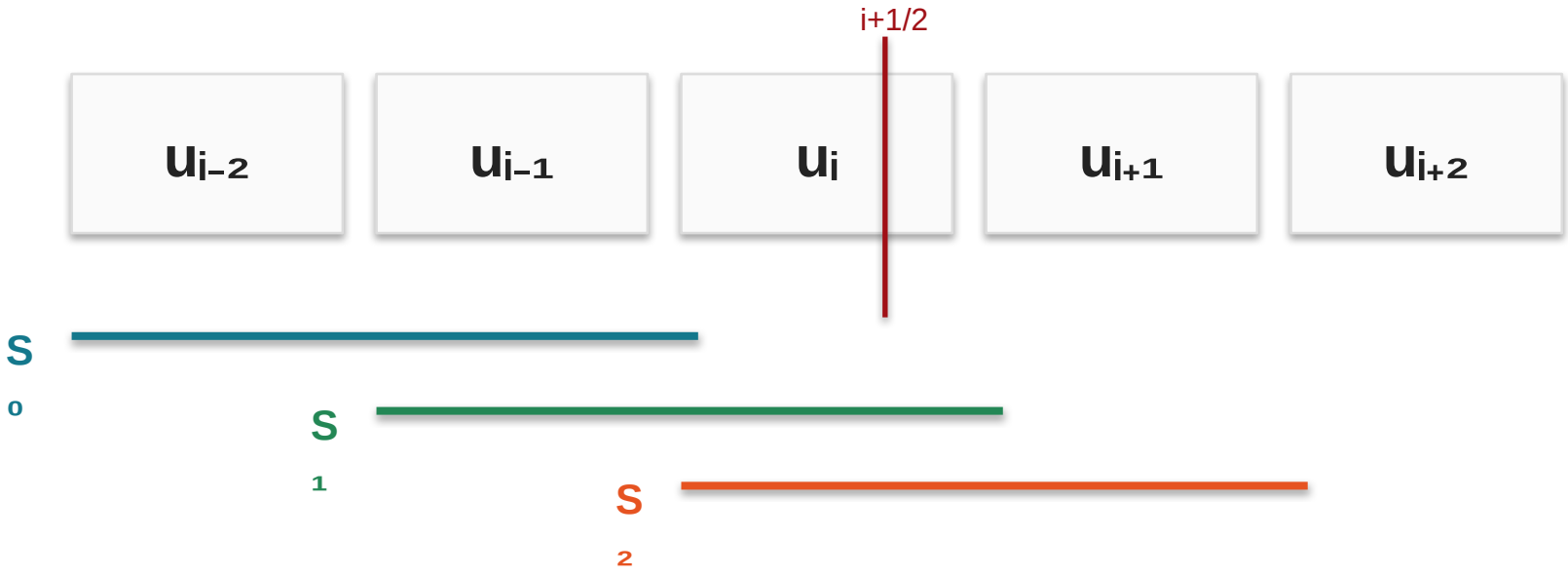
Computational pipeline

1. Start from a conservation law
2. Discretize the local stencil
3. Reconstruct interface values
4. Apply Lax–Friedrichs flux splitting
5. Advance in time with SSPRK3
6. Replace or improve decisions with ML

This single figure links the physical phenomenon, discretization, reconstruction, fluxes and time integration.

Classical reconstruction: how do we reconstruct at interfaces?

At each interface, a 5-point stencil provides three candidate 3-point substencils.



ENO selects
Choose one smooth substencil and ignore the others.

WENO weights
Combine all candidates with nonlinear adaptive weights.

TENO discards
Reject troubled substencils, then renormalize the rest.

Message: ENO selects, WENO weights, and TENO discards.

ML approach: the network receives a local vector and predicts stencil decisions

Input vector fed to the neural network



A 5-point local stencil centered around the target interface.

Feed-forward classifier



5 → 64 → 32 → 16 → 8 → 3

Outputs

$(p_0, p_1, p_2) = \mathrm{NN}_\theta(S_i)$

Probabilities for S_0, S_1, S_2

Interpretation

The network does not solve the PDE directly.
It learns which local stencil information is reliable near discontinuities.

Role in NN-TENO5

The classical keep/discard rule is replaced by learned stencil-decision probabilities.

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Main ML methodology: imitation learning vs unrolled physical training

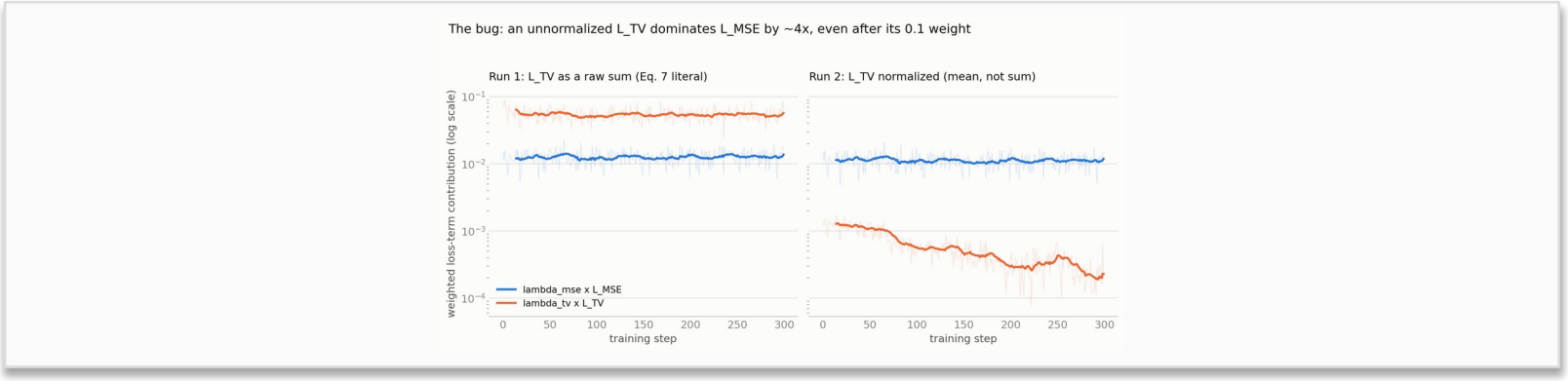
The goal is not only to imitate TENO decisions, but to improve the evolved physical solution.

Path 1 — arni

Stencil vector → Neural network → Predicted mask → TENO labels → Classification loss

Path 2 — Unrolled inin

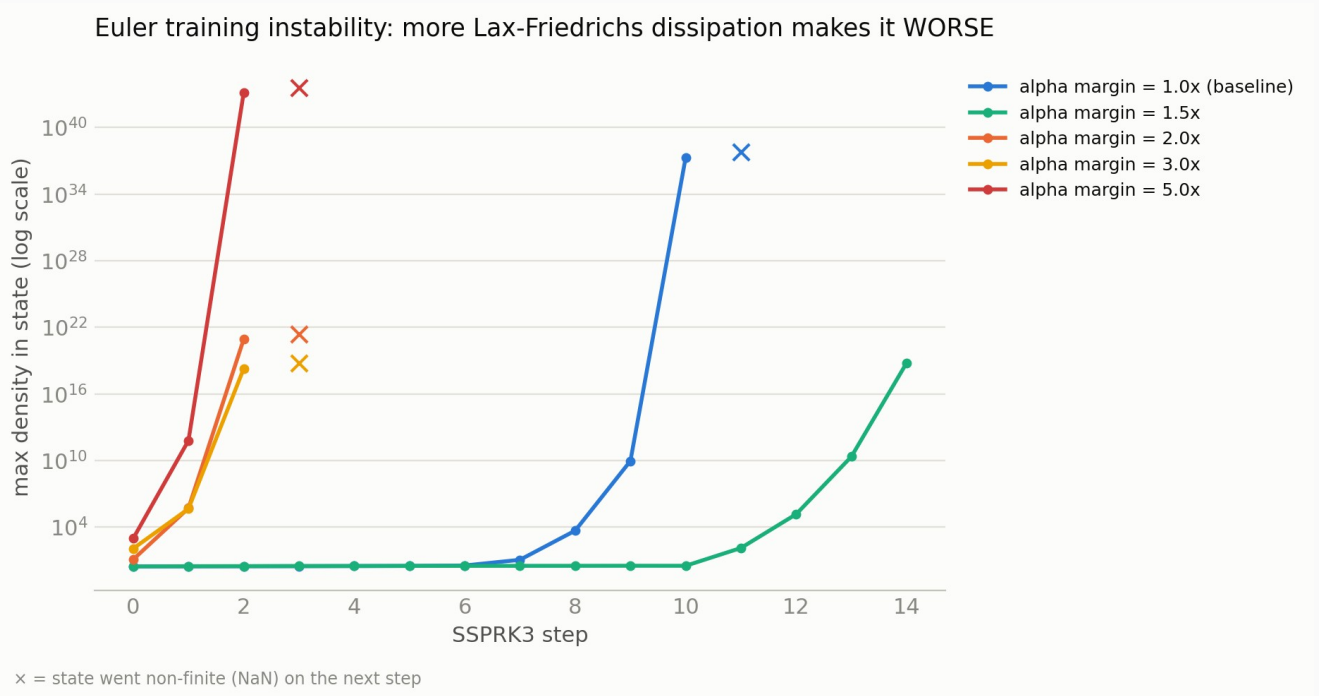
Stencil vector → Neural network → Reconstructio n → Flux splitting → SSPRK3 rollout → Loss vs reference



Real training diagnostic: normalizing the TV term was necessary so that the physics loss became meaningful and balanced.

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Stable training required hybrid and positivity-aware constraints



$$\Phi_{\text{final}} = (1-\lambda)\Phi_{\text{TENO5}} + \lambda\Phi_{\text{NN}}, \quad 0 \le \lambda \le 1$$

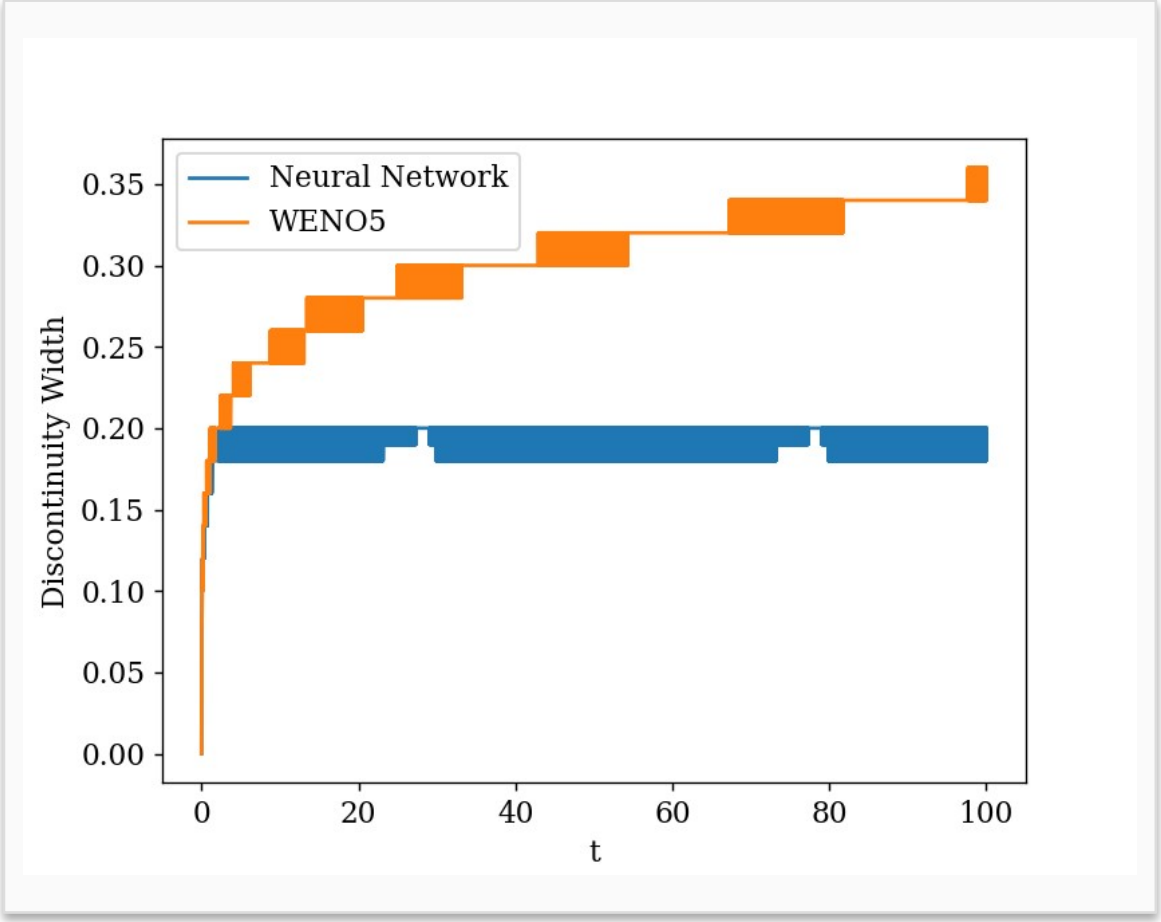
Pure learned reconstructions became unstable for Euler rollouts.

A convex mixing head blends NN output with TENO5.

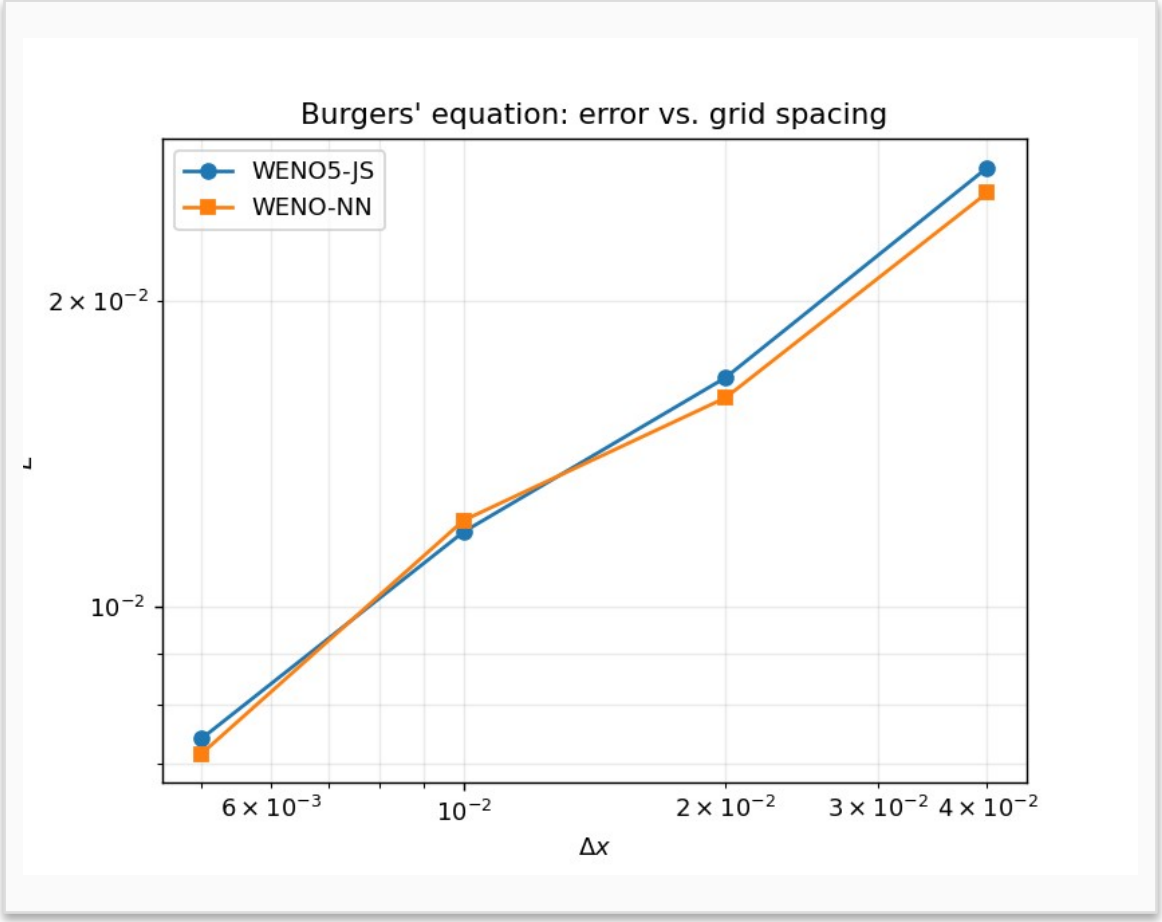
This preserves robustness while keeping learned corrections.

Key lesson: ML must respect the numerical structure and physical constraints of the solver.

Real scalar results from the project



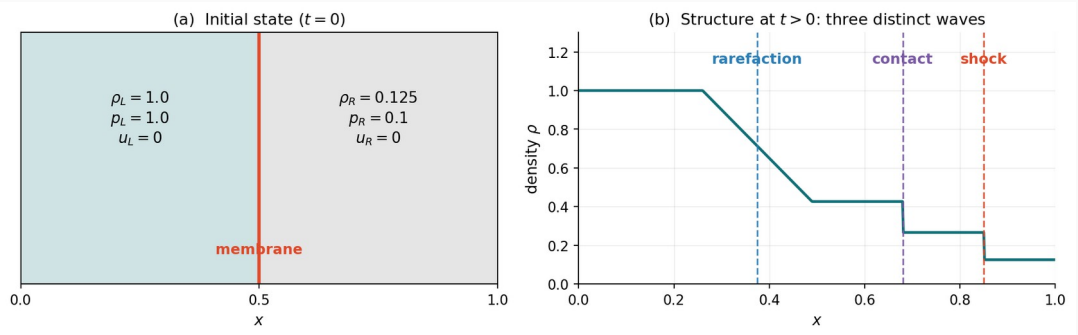
Discontinuity width comparison: Neural Network vs WENO5

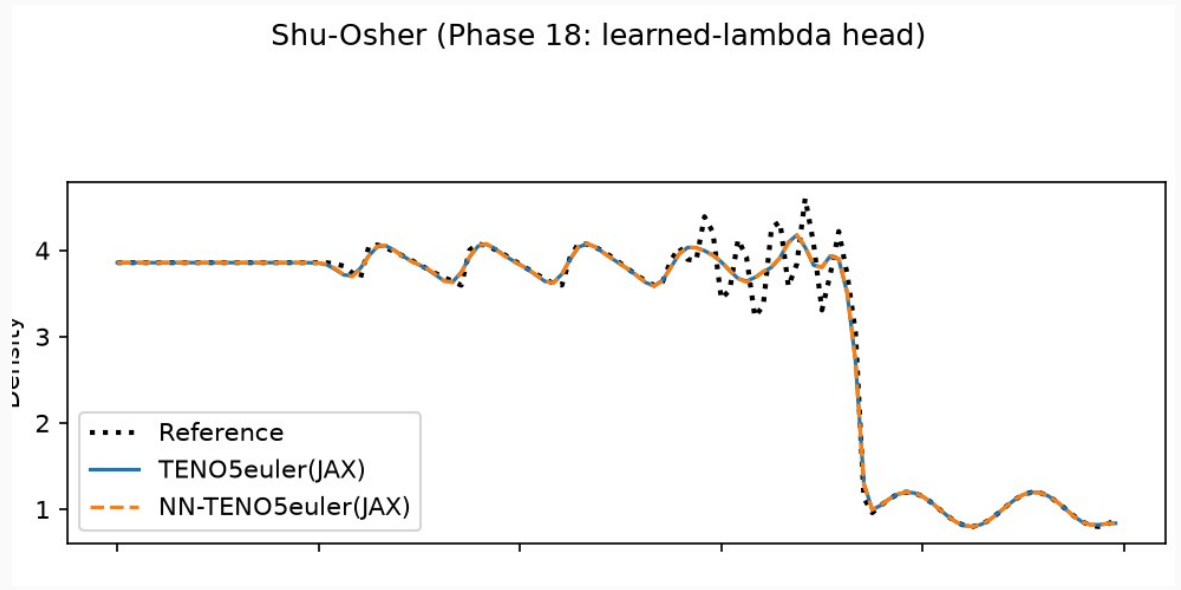


Burgers equation: error vs grid spacing

These real project plots show sharper discontinuities and competitive accuracy on scalar problems.

Euler benchmarks: physical wave structure and final learned result





Sod remains close to parity with classical TENO5.

Shu–Osher shows a modest improvement with the learned mixing head.

For Euler equations, gains are more benchmark-dependent.

Left: Sod wave structure. Right: final Shu–Osher result from the project.

Conclusion

What I built

A framework around WENO/TENO shock-capturing schemes.

Neural enhancements for reconstruction decisions.

Differentiable training with physics-based losses in JAX.

What I learned

Conservation laws and high-order numerical reconstruction.

Flux splitting and SSPRK time integration.

Why ML must respect stability and physical structure.

Take-home message

Learning can improve shock-capturing schemes,
but only inside a robust numerical framework.

Future work

Stronger Euler performance and broader validation.

Extension to 2D and more realistic CFD settings.

More rigorous comparisons with classical baselines.

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Backup — key mathematical ingredients

Conservation law

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0$$

Finite-volume update

$$\overline{u}_i^{n+1} = \overline{u}_i^n - \frac{\Delta t}{\Delta x}\left(\widehat{f}_{i+1/2} - \widehat{f}_{i-1/2}\right)$$

Stencil probabilities

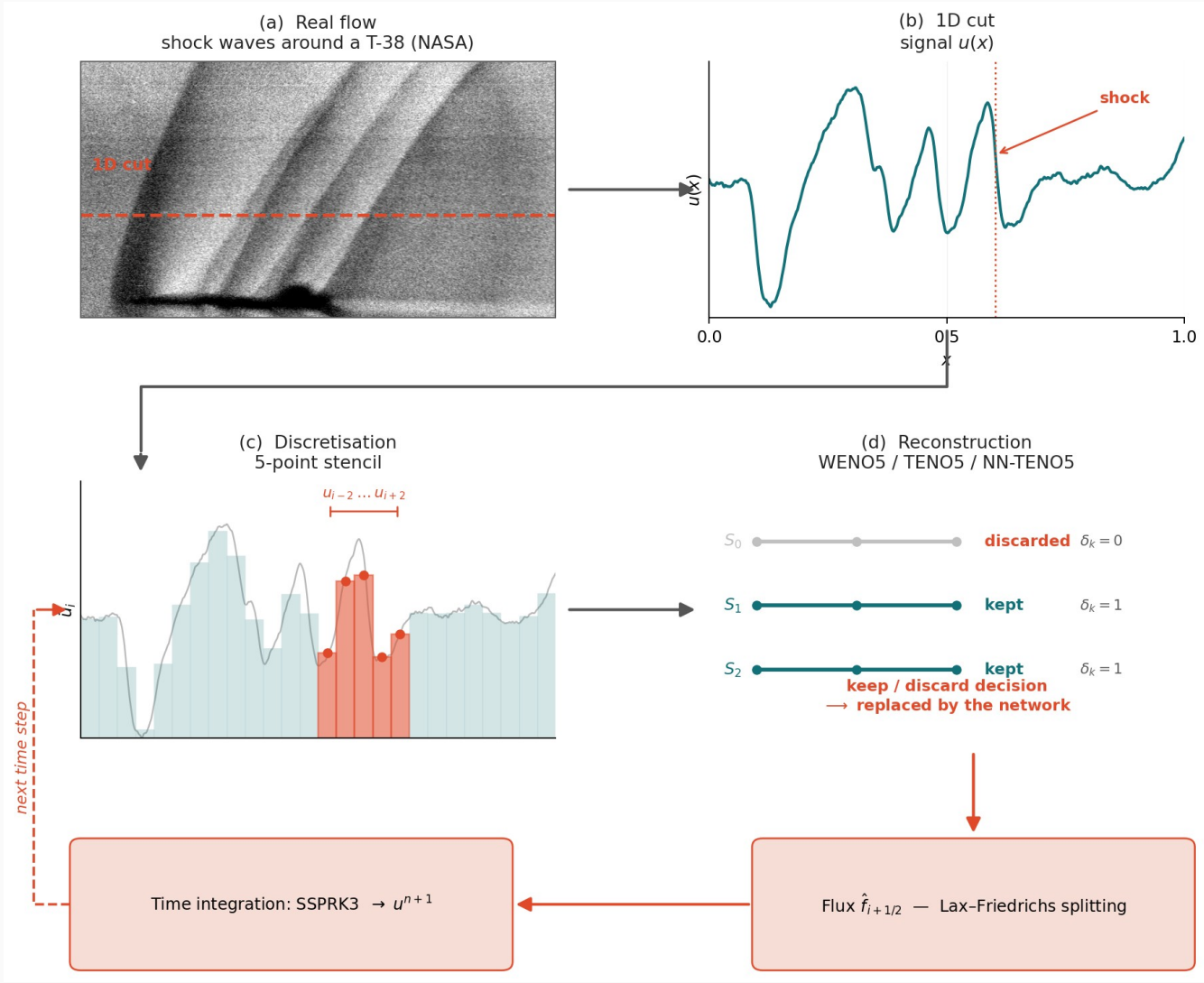
$$(p_0, p_1, p_2) = \mathrm{NN}_\theta(S_i)$$

Hybrid convex mixing

$$\Phi_{\text{final}} = (1-\lambda)\Phi_{\text{TENO5}} + \lambda\Phi_{\text{NN}}, \quad 0 \le \lambda \le 1$$

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Backup — end-to-end project workflow



How the pieces connect

Conservation law

Stencil discretization

Reconstruction

Flux splitting

SSPRK3 time integration

ML-enhanced decision stage

This slide is useful during Q&A if you are asked to summarize the full numerical–ML pipeline.